

CS454 Midterm 1 Part B

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Abstract—The exam is open book. No computers are allowed.

I. QUESTION (CHAPTER 19)

Let us say we have L classification algorithms. How can we order these L from best to worst? (Hint: Use tests comparing two classification algorithms)

II. QUESTION (CHAPTER 6)

Given the following error rates for a four feature problem, what will be the result of the following attribute subset selection procedures?

- Stepwise forward selection
- Stepwise backward elimination

Error Rates: (F_1) , 0.23; (F_2) , 0.25; (F_3) , 0.36; (F_4) , 0.24; (F_1, F_2) , 0.18; (F_1, F_3) , 0.19; (F_1, F_4) , 0.20; (F_2, F_3) , 0.16; (F_2, F_4) , 0.29; (F_3, F_4) , 0.22; (F_1, F_2, F_3) , 0.39; (F_1, F_2, F_4) , 0.45; (F_1, F_3, F_4) , 0.24; (F_2, F_3, F_4) , 0.36; (F_1, F_2, F_3, F_4) , 0.59.

III. QUESTION (CHAPTER 8)

A linearly separable 2-dimensional dataset with 2 classes is given. Propose a condensed 1 nearest neighbor algorithm for this dataset.

IV. QUESTION (CHAPTER 3)

In many pattern classification problems one has the option either to assign the pattern to one of K classes, or to reject it as being unrecognizable. If the cost for rejects is not too high, rejection may be a desirable action. Let

$$\begin{aligned}\lambda(\alpha_i|C_j) &= 0 \quad i = j \\ \lambda(\alpha_i|C_j) &= \lambda_r \quad i = K + 1 \\ \lambda(\alpha_i|C_j) &= \lambda_s \quad \text{otherwise}\end{aligned}$$

where λ_r is the loss incurred for choosing the $(K + 1)$ action, rejection, and λ_s is the loss incurred for

making any error. Show that the minimum risk is obtained if we decide C_i if $P(C_i|x) \leq P(C_j|x)$ for all j and if $P(C_i|x) \leq 1 - \frac{\lambda_r}{\lambda_s}$, and reject otherwise.

V. QUESTION (CHAPTER 5)

Assume that you have a 2 dimensional 2 class dataset. For the following algorithms, give an example estimated covariance matrix S (or matrices S_i depending on the algorithm). Also plot the covariance matrices.

- Quadratic Discriminant Analysis
- Linear Discriminant Analysis
- Naive Bayes
- K-means

VI. QUESTION (CHAPTER 4)

Let x have a Rayleigh density, for all $x \geq 0$

$$p(x|\theta) = 2\theta x e^{-\theta x^2}, \quad (1)$$

Suppose that n samples $x_1, x_2, \dots, x_n$ are drawn independently according to $p(x|\theta)$. Show that the maximum likelihood estimate for θ is given by

$$\hat{\theta} = \frac{n}{\sum_{k=1}^n x_k^2} \quad (2)$$

VII. QUESTION (CHAPTER 7)

Suppose that the task is to cluster the following eight points (with (x, y) representing location) into three clusters. $A_1(2, 10)$, $A_2(2, 5)$, $A_3(8, 4)$, $A_4(5, 8)$, $A_5(7, 5)$, $A_6(6, 4)$, $A_7(1, 2)$, $A_8(4, 9)$. The distance function is Euclidian distance. Suppose initially we assign A_2 , A_5 , and A_8 as the center of each cluster, respectively. Run the k-means algorithm one epoch on this data.